Subset Sums in a Fourfold Sumset

Anchored hole bounds and integer center bases,
and the audited bound $N \le 91$

Sum-of-Subset research project

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Abstract

We continue the audited bound $N \le 92$ by excluding cardinality 92. The former uniform inverse margin becomes negative in fourteen orientations. A progression/non-progression dichotomy repairs this gate: non-progressions improve both the support and defect budgets; two progression supports are treated exactly by residue classes. The only unequal-step case left by the cardinality budget contradicts the subsequent peeling budget. We then use the persistent-hole theorem earlier, after core normalization but before propagating the lattice. A full-prefix central interval lowers its density threshold by one; a sharp non-progression anchor strengthens the defect recurrence. For the size-92 model, these reduce the high-hole covering cap from 199 to 150. For covering, integer row operations on actual contributing centers replace the former one-coordinate propagation. All three-empty modes become integral without assuming contribution by the leading translates. Four empty units now occur; all 10,880 modes are retained, with partial integer information in the 30 exceptional modes. Finally, the second core cutoff supplies stronger leading quantiles. A cardinality charge after trimming one or both extremes strengthens the twofold-sumset bound without classifying individual core sets. Low-hole mass transport also retains the stronger support-excess floors from the hole witnesses. Exact integer certificates, a standard-library checker, and complete run records accompany the report. The stronger archived bound 84 remains pending complete independent replay.

1 Result and inherited structure

For a finite real set A, let $\mathrm{FS}(A)$ include the empty sum and let $4A = A + A + A + A$ allow repeated summands. Write N for the largest feasible cardinality under $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 91 elements; hence* $71 \le N \le 91$.

The unchanged lower construction is $([-31, 31] \cap \mathbb{Z}) \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The audited foundation $N \le 92$ is [2]. It suffices to exclude size 92. Such a set contains zero, since adjoining zero to one without it preserves feasibility and would give 93 elements. No deletion principle for arbitrary subsets is assumed.

We inherit the corrected integralization and peeling lemmas [6], the sharp four-extreme and joint-covering bounds [5], the threshold-17 non-progression theorem [4], and the exact prefix inverse [3]. For a positive r-set,

$$T_r = \frac{r(r+1)}{2}, \quad \mu(X) = |\,\mathrm{FS}(X)| - 1 - T_r, \quad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where δ_0 counts x for which $2x$ is not a distinct-element two-sum of $X \cup \{0\}$. The maximum is defective, so $\rho \le \mu$. The general floors are $\rho \ge -8$ and $\rho \ge 0$ for $r \ge 14$. For non-progression support, $\mu \ge r - 3$, $\rho \ge r - 14$ for $r \ge 6$, and $\rho \ge r - 5$ for $r \ge 17$.

Peel as before to $D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$, with

$$d = 84, \quad e = 76, \quad a + b = m = 75, \quad I = (a - 37)(a - 38), \quad \Lambda = 1445 + I, \quad \Omega = 500 - I.$$

Put $\rho = \rho(P) + \rho(Q)$ and $\mu = \mu(P) + \mu(Q)$. The exact inherited identity is

$$\rho + 2\vartheta + c + \mathcal{T} = \Omega, \qquad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g, \qquad c = 2|2D| + 18d + 85 - |\operatorname{FS}(D)|. \tag{1}$$

All displayed slacks other than ρ are nonnegative, and $\kappa = |\operatorname{FS}(P) - \operatorname{FS}(Q)| - |\operatorname{FS}(P)| - |\operatorname{FS}(Q)| + 1$. The deficient-sign inequality is $1 + T_{76} > 24 \cdot 92 + 77$. The paired bootstrap gives $q^2 \le 2593$ with odd q, hence $q \le 49$ and $a, b \ge 13$. At size 13 on one side, $\Omega = -100 < -8$; the other side has size 62 and nonnegative defect. This is impossible. Both sides are now at least 14; $\Omega \ge 0$ leaves $16 \le a \le 59$.

2 Repairing the simultaneous inverse gate

The sufficient inverse condition used in the foundation is

$$\kappa < \min(|\operatorname{FS}(P)|, |\operatorname{FS}(Q)|) - 3. \tag{2}$$

Let $r = \min(a, b)$, $t = \max(a, b)$, and $J = T_r - 2 - \Omega$. Its minimum is now -21. The fourteen orientations with $J \le 0$ have $22 \le r \le 28$.

Lemma 2.1 (Use non-progression before the inverse theorem)**.** *In each exceptional orientation,* (2) *holds whenever at least one side-support is not a progression.*

Proof. If the side of size r is non-progression, its support grows by at least $r - 3$ above $1 + T_r$ and its defect is at least $r - 5$. Since $t > r$, $T_t \ge T_r + r - 3$. Thus the margin in (2) is at least $J + 2r - 8$. If only the larger side is known to be non-progression, its defect gives margin at least $J + t - 5$. Here we used $\kappa \le \Omega - \rho$, directly from (1). The two exact minima over the fourteen orientations are 20 and 24. □

The exceptional sizes, with both orientations included, are as follows. The last column is the residual budget in the double-step progression case treated below.

r	Ω	J	$J + 2r - 8$	$J + t - 5$	$\Omega - T_r$
22	260	-9	27	39	7
23	290	-16	22	31	14
24	318	-20	20	26	18
25	344	-21	21	24	19
26	368	-19	25	25	17
27	390	-14	32	29	12
28	410	-6	42	36	4

Lemma 2.2 (Two progression supports)**.** *Let $U = p[0, u-1]_{\mathbb{Z}}$ and $V = q[0, v-1]_{\mathbb{Z}}$, with $p, q \ge 1$,* $\gcd(p, q) = 1$*, and $u, v \ge 2$. Then*

$$|U + V| = \begin{cases} uv, & p \ge v \text{ or } q \ge u, \\ pu + qv - pq, & p < v \text{ and } q < u. \end{cases}$$

Proof. A collision requires index differences divisible by q and p, respectively. There are none in the first case. In the other case, group indices of U by $i = i_0 + qk$, $0 \le i_0 < q$. The corresponding sums in their residue class are translates of $[0, v-1]$ by successive multiples of p. These intervals overlap because $p < v$. If there are $K + 1$ indices in the class, its union has $v + pK$ values. Summing K over the q classes gives $u - q$, and the asserted cardinality follows. □

Suppose now both supports are progressions in an exceptional orientation. Integralize first, divide their steps by their gcd, and label the side of size r as U, so $u \ge 1 + T_r$ and $v \ge 1 + T_t$. Reflection of the second support does not change the sumset cardinality. The collision-free case gives $\kappa \ge T_r T_t > \Omega$. In the other case,

$$\kappa = (p-1)(u-q) + (q-1)(v-1).$$

Since $T_t > \Omega$, we must have $q = 1$. Then $\kappa = (p-1)(u-1)$, and $2T_r > \Omega$ forces $p \le 2$. If $p = 1$, both supports already have a common step and no hull holes; no inverse theorem is needed.

It remains to exclude $p = 2, q = 1$. The difference support is a full interval because $v \ge 2$. On this common lattice,

$$h_P + h_Q = \kappa = u - 1, \qquad z = 0, \qquad \Omega - \kappa \le \Omega - T_r \le 19.$$

This case has $h_U = |U| - 1$, so we do *not* apply any theorem requiring a strict side-hole bound. Instead use the general span lemma $\max X \le |X| + \rho(X) + h_X + 2$, valid for every positive integer set of size at least 15. For completeness, delete its largest element $r+q$ and write X' for the remaining set. If $\nu \ge 0$ counts destroyed diagonal defects, the exact increment identity is $\rho(X) = \rho(X') + q - h_X + h_{X'} - 2 + 2\nu$. Since $|X'| \ge 14$, all terms other than $q - h_X - 2$ are nonnegative. This proves the stated span bound without a primitivity hypothesis. The two lattice-propagation comparisons below apply directly, with $z = 0$. They put D_1, D on this lattice and propagate zero holes. The sharp four-extreme theorem then gives

$$\mathcal{T} \ge \kappa + 158,$$

contradicting (1) and $\Omega - \kappa \le 19$. The strict contradiction has margin at least 139. This excludes every unequal-step progression case left by the budget. Lemma 2.2 is also checked against 8,428 directly enumerated small progression pairs; its arbitrary-length proof is the residue argument above.

3 An earlier use of persistent holes

In the remaining cases, the inverse theorem or equal progression steps give a common core lattice with

$$h_P, h_Q \le \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \le z \le \min(h_P, h_Q), \quad h_X < |\operatorname{FS}(X)| - 1.$$

The inverse is not reapplied to the excluded double-step case. The persistent-hole argument can be shortened and strengthened in two places. Apply the central-interval theorem to the full prefix, avoiding the deletion of its maximum. Then start the defect recurrence at the sharp positive floor of the preceding non-progression prefix.

Lemma 3.1 (Full-prefix endpoint holes)**.** *Let $Y \subseteq [1, i+\delta]_{\mathbb{Z}}$ have i elements, where $0 \le \delta \le \beta$ and $i \ge 2\beta + 4$. With $S = \sum Y$,*

$$\operatorname{FS}(Y) = [0, S]_{\mathbb{Z}} \setminus (H \cup (S - H)), \qquad H \subseteq [1, 2\beta], \qquad |H| \le \beta.$$

Proof. Theorem 1 of Lev [1] states that an i-element subset of $[1, l]_{\mathbb{Z}}$ with $l \le 3i/2 - 2$ has all subset sums in $[2l - 2i + 1, S - (2l - 2i + 1)]_{\mathbb{Z}}$. Here $l = i + \delta$ meets that hypothesis, so the interval $[2\delta + 1, S - 2\delta - 1]$ is full. Every missing lower sum in $[1, 2\delta]$ is a missing member of Y, because $2\delta \le i + \delta$; there are at most δ such members. Complement symmetry determines the upper holes. Enlarging the endpoint range from 2δ to 2β proves the statement. Unlike the previous counted-endpoint argument, this use does not need to identify the number of deletions with the exact support increment. □

Theorem 3.2 (Anchored persistent-hole bound)**.** *Let X be a positive integer r-set with $2\beta < h_X < |\operatorname{FS}(X)| - 1$, where $\beta \ge 2$. Choose $\varepsilon \in \{0, 1\}$ and $r \ge L \ge \max(17 + \varepsilon, 2\beta + 4)$. Then*

$$\rho(X) \ge B_\varepsilon(r, \beta, L) := \min \begin{Bmatrix} \varepsilon(L-6) + (\beta - 1)(r - L + 1), \\ T_{L-\varepsilon} - r - 1, \\ T_L + L - 6\beta - r - 7 \end{Bmatrix}. \tag{3}$$

Proof. Assume ρ_r is less than all three terms. The strict upper hole condition forces X to be primitive: a gcd at least two creates at least $|\operatorname{FS}(X)| - 1$ hull holes. Its support is non-progression because a primitive progression support containing zero is a full interval. For the increasing prefixes

X_i, put $y_i = |\operatorname{FS}(X_i)| - |\operatorname{FS}(X_{i-1})| - i$. The inherited increment identity gives $\rho_i \geq \rho_{i-1} + y_i - 2$; for $i \geq 6$, $y_i = 0$ forces progression support.

If the last progression prefix has size $t \geq L - \varepsilon$, its next support translate is disjoint. Indeed, an on-lattice translate that meets or abuts the previous progression preserves progression support; an off-lattice translate is disjoint. Thus $y_{t+1} \geq T_t - t$, and all subsequent increments are at least one. Using $\rho_t \geq 0$ gives

$$\rho_r \geq T_t - t - 2 - (r - t - 1) = T_t - r - 1 \geq T_{L-\varepsilon} - r - 1,$$

a contradiction. All prefixes from $L - \varepsilon$ onward are therefore non-progressions. If $\varepsilon = 1$, the sharp theorem at size $L - 1 \geq 17$ gives $\rho_{L-1} \geq L - 6$; if $\varepsilon = 0$, the general floor gives $\rho_{L-1} \geq 0$. If $y_i > \beta$ throughout $L \leq i \leq r$, telescoping from this anchor gives the first term of (3), again impossible. Choose $i \geq L$ with $1 \leq y_i \leq \beta$.

Since $i \geq 2\beta + 4 \geq \beta + 5$, the inherited small-increment inverse applies and writes $X_i = tY$ with Y primitive, $|Y| = i$, and $\max Y \leq i + \beta$. The scale t is an integer since X_i is integral and $\gcd(Y) = 1$. If $t > 1$, final primitivity forces a first later element off $t\mathbb{Z}$, say at index k. Its support translate is disjoint. The same increment calculation, with $\rho_{k-1} \geq 0$, gives $\rho_r \geq T_{k-1} - r - 1 \geq T_L - r - 1$, contradicting the second term. Hence $t = 1$ and Lemma 3.1 applies to X_i.

Suppose the endpoint-hole description holds at size $v \geq i$, with sum S, and let x be the next element. If $x \geq S - 4\beta$, its translate adds at least $S - 6\beta$ points above S. For $x \leq S$, at least $x - 2\beta$ of the last x positions are present; for $x > S$, the whole translate is new. Consequently $y_{v+1} \geq S - 6\beta - v - 1$. Using the sharp floor $\rho_v \geq v - 5$ and all later increments at least one gives

$$\rho_r \geq S + v - 6\beta - r - 7 \geq T_L + L - 6\beta - r - 7,$$

contradicting the third term. Therefore $x < S - 4\beta$. Also $x > 2\beta$ since x exceeds the maximum of a prefix of size at least $2\beta + 4$. The two central intervals overlap or abut; the lower holes remain H, and complement symmetry supplies the upper holes. This propagates to X_r, giving $h_X \leq 2\beta$, the final contradiction. □

For $z \geq 19$, $(\beta, L, \varepsilon) = (9, 22, 1)$ gives respective floors $136, 152, 168$ at sizes $36, 38, 40$, instead of the previous $112, 128, 144$. The new helper `verification/nonap_hole_rho_full.py` checks 22,335 parameter rows, retaining both values of ε. It also checks the endpoint description directly in 88,392 deleted intervals at the boundary $i = 2\beta + 4$, for $2 \leq \beta \leq 6$ and $0 \leq \delta \leq \beta$. These finite checks supplement the general proof; they do not substitute for the central-interval theorem.

Let $\mathcal{R}(r, z)$ be the maximum of zero and these accepted floors with $2\beta < z$, $\beta \leq 20$, both permitted values of ε, and all permitted L. Let $F(r, u)$ be the unchanged exact prefix inverse of [3], and write $W(r, u, z) = \max\{F(r, u), \mathcal{R}(r, z)\}$.

Put $C = 2m + 4e + 14 - \Lambda = -977 - I$. The inherited inner comparisons are

$$|\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| \leq \min\{C + \Omega + \rho - \mu + 2z,\ C + 3\Omega - \rho - \mu\}.$$

For $z \leq 14$, $\rho \leq \mu$ gives the upper bound $-449 - 2I$. For $z \geq 15$, first use $\mu \geq 69$ and the applicable non-progression floors. The only unresolved sizes have $a, b \geq 28$. The inherited effective-hole witnesses at $\beta = 7$ supply lower bounds $u_r = r + C_7(r) + 1$ for $\mu(X)$; now use

$$\rho + \mu \geq u_a + u_b + W(a, u_a, 15) + W(b, u_b, 15).$$

The high-hole inner comparison is at most -31 over all 44 orientations. This use of (3) is legitimate because P, Q have already been normalized, even though the peeled layers are not yet known to lie on their lattice. The outer comparison is

$$\Lambda - (4d - 10 + \Omega) = 619 + 2I > 0.$$

Disjoint translates would violate these strict factor-two comparisons, so D_1, D lie on the core lattice. In the double-step progression case of the previous section only the low-hole comparison is used; its derivation needs the general span lemma, not a primitive side support.

Hole propagation and the sharp four-extreme theorem now give

$$w \le 4z, \quad \mathcal{T} \ge \kappa + 158 - 3z, \qquad \mathcal{T} \ge \max(53, z, 158 - 2z). \tag{4}$$

Neither size-16 boundary can survive these budgets. Thus all subsequent calls to F and $\mathcal{R}$ are within their proved range $r \ge 17$.

4 Integer bases of contributing centers

Write the original four positive outer magnitudes as $p_1 > p_2 > p_3 > p_4 > M_p$, and similarly use $q_i > M_q$ on the negative side. The core lattice already contains M_p, M_q and all of D. Set

$$c_p = p_1 - p_2 - p_3 - p_4, \qquad b_p = -p_1 + 2p_2 - p_3 - p_4,$$

and reflect for q. The twenty ordinary center forms are

$$\begin{aligned} v_{pi} &= c_p + p_{i+1}, & v_{p,i+4} &= c_p - q_{i+1} & (0 \le i < 4),\\ v_{p8} &= p_2 - p_3 - p_4, & v_{p9} &= b_p, \end{aligned}$$

and their reflected forms. The eighteen-unit selection and the two branches $b_p + b_q \le 0$ and $b_p + b_q > 0$ are inherited from [5, 2]. Every actual contributing ordinary center is integral: its translate of the integral set D intersects the integral target. The actual member of a split pair is treated in the same way.

Lemma 4.1 (Integer row certificates)**.** *Let M have, as rows, the coefficient vectors of known integral center forms in the eight outer coordinates. If an integer row vector v is an integer combination of rows of M, then $v \cdot (p, q)$ is integral. In particular, an integer matrix U with $UM = I_8$ proves integrality of all eight outer coordinates.*

Proof. Integer linear combinations of integers are integers. This also proves the claim for individual gaps and for the branch form $b_p + b_q$. □

The new helper `verification/integral_centers.py` uses row swaps, sign changes, and integer row subtraction to construct an echelon basis by the Euclidean algorithm. It tracks the integer combinations and checks $H = UM$ independently by matrix multiplication. A target form is accepted only when integer elimination against H leaves zero remainder. A nonunit pivot is never divided away. Thus row reduction is a means of finding explicit integer combinations, not a rational rank test.

For zero through three empty units, the ordinary contributing forms alone recover all coordinates in respectively $4, 70, 578, 2992$ modes. Leading contribution is not needed for these assertions. For example, the old exception in branch B with missing $p0, p1, p8$ and second-pair choice p has

$$p_2 = c_p + v_{p9} + 2p_3 + 2p_4.$$

The retained q centers recover its side, $v_{p4} + q_1$ recovers c_p, and the retained own centers recover p_3, p_4. The displayed integer combination then recovers p_2, followed by p_1. This is the combination missed by the former one-unknown rule.

Four empty units are now possible. All 10,880 modes are enumerated. Ordinary centers recover 10,812 of them. If the residual cap is below $2d - 1 = 167$, a leading center may be added *only on a side whose own-center block is nonempty.* That block charges the actual leading loss, which is then less than $|2D|$ and proves contribution. This recovers 38 more modes, for a total of 10,850. The remaining 30 modes are not discarded: 12 have rank seven, and 18 have rank eight with a nonunit obstruction. The checker retains every individually proved integral gap or branch form and leaves other strict gaps weak in the real relaxation. Their integer-gap counts are

$$6 \text{ modes with } 4, \quad 18 \text{ with } 5, \quad 4 \text{ with } 6, \quad 2 \text{ with } 7.$$

No leading quantile is imposed on a side with no charged own-center block.

5 A deeper leading quantile

Retain C_p, the second-largest element of the positive core P, and C_q on the other side. The mass and independent-cutoff rows of [2] remain valid. With $x_{-1} = C_p, x_0 = R_p, x_4 = M_p$, and similarly y_i, these include

$$C_p \geq a-1, \quad C_p + 1 \leq R_p, \quad C_p + R_p + x_1 + x_2 + x_3 + M_p - S_p \leq -T_{a-2},$$
$$K := |2D| \leq M_p + M_q + x_i + y_h + 1 + T_{4-i} + T_{4-h} \qquad (-1 \leq i, h \leq 4).$$

The 36 tail bounds are inherited, not claimed as a new theorem here. The new use of C_p, C_q concerns the leading translates.

Lemma 5.1 (Leading quantiles from a smaller core)**.** *Let a leading center c target $[1, S]_{\mathbb{Z}}$ and let L be its actual total loss from $c + 2D$. If $L < 2d - 11$, then*

$$c + 2C_p + L \geq 2d - 11, \qquad c - S - 2C_q - L \leq 12 - 2d.$$

For $d = 84$, the two right sides are 157 and -156.

Proof. Remove the five largest positive elements of D. The remaining set has $d - 5$ elements, maximum C_p, and a twofold sumset of at least $2d - 11$ points. Its translate is a subset of $c + 2D$. Since its total loss is at most $L < 2d - 11$, it contributes to $[1, S]$. At most $c + 2C_p$ positive integers lie at or below its maximum; at most L further points lie below 1. This proves the first inequality. Remove the five most negative elements instead. The minimum is $-C_q$. Counting integers from $c - 2C_q$ to S, with at most L points above S, proves the second inequality. □

The proof uses the same total loss in two alternative necessary inequalities; it does not add the losses twice. In the model the rows are enabled only when the entire group's residual cap is less than 157 and that side's leading loss is charged. They are used only in the nested model, where C_p, C_q are retained.

6 Trim first, then bound the sumset

The independent hull cutoffs can overcount the contribution of a widely separated extreme. Its number of possible partners gives a second, independent upper bound.

Lemma 6.1 (Cardinality cost of trimmed extremes)**.** *Let D have d elements and $F = D \setminus T$, where $|T| = s$. Then*

$$|2D| \leq |2F| + sd - T_{s-1}.$$

Proof. Use $2D = 2F \cup (F + T) \cup 2T$, with $|F + T| \leq s(d - s)$ and $|2T| \leq s(s + 1)/2$. The sets may overlap; summing their cardinalities is an upper bound. The last two terms sum to $sd - T_{s-1}$. □

Remove the k largest positive and the ℓ most negative elements, where $(k, \ell) \in \{(1, 0), (0, 1), (1, 1)\}$. Put $A = 4 - k$, $B = 4 - \ell$, and $s = k + \ell$. The remaining extremes are $x_A, -y_B$ in the preceding notation. The same independent-cutoff proof applied to F gives, for $-1 \leq i \leq A$ and $-1 \leq h \leq B$,

$$K \leq x_A + y_B + x_i + y_h + 1 + T_{A-i} + T_{B-h} + 84s - T_{s-1}. \tag{5}$$

There are $30 + 30 + 25 = 85$ such rows. They do not require the original outer coordinates to be integral, and are valid for every actual D. All mass-transport coefficients in these rows are zero. They are retained in nested models, while coarse models may discard them.

For illustration, a size-92 relaxation in the four-empty family had $R_p = 41$, $R_q = 38$, $x_3 = 50$, $y_3 = 47$ and allowed $K = 373$. Removing both extremes and using $i = h = 0$ gives instead

$$K \leq 50 + 47 + 41 + 38 + 1 + 6 + 6 + 167 = 356.$$

This is an excluded relaxation point, not a feasible construction. The proof no longer needs an enumeration of small-defect core sets to remove this obstruction.

The checker directly enumerates the same 784 small set pairs used for the inherited core-cutoff audit. All 66,640 trimmed cutoff checks pass; 735 of these pairs have a strictly better minimum bound than the 36 untrimmed cutoffs. This finite check supplements Lemma 6.1 and the general cutoff proof. The three-element example $D = \{-3, 0, 1\}$ shows why the two-extreme increment cannot be reduced from $2d - 1$ to $2d - 2$: $|2D| = 6$ and $|2\{0\}| = 1$.

7 Complete size-92 model universe

Let $U(r, z)$ be the inherited effective-hole lower bound for μ, with $U(r, 1) = U(r, 2) = r - 3$. For positive holes the scalar cap is

$$c \le 500 - I - W(a, U(a, z), z) - W(b, U(b, z), z) - \max(53, z, 158 - 2z).$$

For zero holes use $c \le 342 - I$. In particular $342 \ge 4d = 336$ but $342 < 5d$; retaining only three empty units would be incomplete. There are 460 surviving low-hole states and 2,968 high-hole states. The positive low bands are

Band	Hole range	Combined defect floor	$c + I$ cap	Endpoint
P	1–2	65	281	2
Q	3–4	77	273	4
R	5–14	100	270	14

The latter two defect floors follow from the exact $\beta = 1, 2$ witnesses and the prefix inverse, over all 42 orientations with $17 \le a \le 58$. We also retain the support excess supplied by those same witnesses. For every $17 \le r \le 58$, the effective-hole parameter checks accept $(\beta, C) = (1, 5)$ and $(2, r - 6)$, respectively. Since the conclusion is $\mu \ge r + C + 1$ when $h_X > 2\beta$, the necessary mass floors are

$$\epsilon_r = \begin{cases} r - 3, & 1 \le z \le 2, \\ r + 6, & 3 \le z \le 4, \\ 2r - 5, & 5 \le z \le 14. \end{cases}$$

All 84 parameter witnesses are checked exactly. The latter two rows use $h_X \ge z$ and the strict side-hole bound established after the inverse gate. In particular, they are not applied to the excluded double-step progression case. The mass inequalities become

$$S_p \ge T_a + \epsilon_a + z + x_1 + x_2 + x_3 + M_p, \qquad 2S_p \ge 2T_a + 2\epsilon_a + 2z + 2M_p + 6R_p + 12,$$

and their reflections. Keeping only the weaker $\epsilon_r = r - 3$ in all three bands admits a residual R_2 relaxation; the stronger floors preserve information already proved by the hole witnesses. Each band now permits three empty units. Before endpoint transport impose its constant cap; then move z to its upper endpoint Z and increase S_p, S_q by $Z - z$. The new mass floors depend only on side size, so this transport remains valid. Every retained row has $v_{S_p} + v_{S_q} + v_z \le 0$, except the upper-hole row, which becomes an equality. This invariant is checked for every rebuilt model.

For $z \ge 15$, the strengthened bound gives $c \le 150 < 2d$. Thus two empty units are impossible, every charged leading loss is below 157, and a one-empty model has residual cap at most $150 - 84 = 66 < 80$. All high groups consequently license the deeper leading quantiles; their one-empty cases also license core quantiles. The actual contributor fixes the direction of the active-center test, as in the preceding reports. The remaining states comprise 1,068 zero-empty and 1,022 one-empty states, counting each permitted empty count separately. They are grouped by $(a, j, U(a, z), U(b, z), v_p, v_q)$, retaining one common hole coordinate across each group's interval. There are 54 groups, with 871,618 source-to-group row checks over 2,090 retained states. With all other conditions fixed, replacing only the previous persistent-hole floor reduces the complete structural universe from 35,372

to 28,180 groups; the high-hole portion drops from 9,916 to 2,724. The comparison is recorded in `hole_bound_comparison.json` in the results directory. The former universe is only a comparison of necessary conditions, not an additional completed certificate. For $E_* = 500 - I - v_p - v_q$ and $\delta = c - 84j$, retain together

$$\delta \le C_* - 84j, \quad \delta + z \le E_* - 84j, \quad \delta \le E_* - 53 - 84j, \quad \delta - 2z \le E_* - 158 - 84j.$$

The hole variable may not be maximized independently of the cap.

The mass rows are the preceding report's rows with $a + b = 75$; the span and cardinality budgets become

$$R_p + R_q + \delta - 4z \le 421 - I - 84j, \qquad S_p + S_q + \delta - 2K - 4z \le 1596 - 84j.$$

The dense-span lemma $2R \ge 2r + \min(z, r - 3)$ is unchanged, including its valid chord on a high group's fixed hole interval. Joint block deficits use the core size $e = 76$ and are charged only once. For a block of k retained ordinary centers,

$$O \ge 76k - (R_p + R_q + v_{\max} - v_{\min} + 1) + \max_v (l_v + u_v).$$

The inherited leading-plus-own-center block inequalities use the same replacement of 77 by 76; no leading loss is charged for an absent block. The coarse bound is $K \le M_p + M_q + R_p + R_q + 21$; nested models retain the five envelope levels, all 36 core cutoffs, the 85 trimmed cutoffs, and six inner-loss positions. The group-wide residual cap C_g controls the quantile rows:

Rows	Required cap
Ordinary quantiles	$C_g < r_{\min} + 5$
Core quantiles	$C_g < 80$
Basic leading quantiles	$C_g < 167$ and a charged own block
Nested leading quantiles	$C_g < 159$ and a charged own block
Second-core leading quantiles	$C_g < 157$ and a charged own block

Global quantiles use 84 and 83, core quantiles 80 and 79, and ordinary quantiles $b + 5$ and $a + 4$ on side p. A leading nested level i uses $159 + 2i$ and $158 + 2i$ for $0 \le i \le 4$. Strict outer gaps and the strict branch-B row are licensed by their individual integer row certificates, not by an empty-count shortcut. All inherited binary alternatives remain available, with both children required and no restriction on how many large-loss centers occur.

8 Exact certificates and reproduction

Every root, matrix and right side is rebuilt from these necessary conditions. The low universe contains 268,432 integer root systems. The complete verified statistics are

Quantity	Count
Structural groups	28,180
Integer root systems	271,156
Coarse / nested groups	27,795 / 385
Symbolic interval checks	28,412
Integer extremum evaluations	74,021
Integer roots requiring trees	1,342
Binary internal nodes / leaves	3,320 / 4,662
Maximum tree depth	14
Final integer leaf systems	274,476
Distinct integer vectors	4,130

For $Ax \le b$ and $x \ge 0$, an integer multiplier $w \ge 0$ with $w^T A \ge 0$ and $w^T b < 0$ is a contradiction. The minimum normalized strict margin is $3/2519$. Symbolic right sides are exact quadratics; their integer extrema are checked at endpoints and around the vertex. The complete integer parameter range is covered exactly once, and every binary tree certifies both children. High-hole certificates cover their common real hole interval simultaneously. The saved multiplier inventory confirms use of the trimmed-sumset rows in 2,787 vectors covering 324 structural groups, and use of the second-core leading quantiles in 39 vectors covering 28 groups. These counts are recorded in `method_usage.json`.

The 36 negative controls include omission of the new Z_4, Q_3, R_3 families, an unjustified raw inverse gate, division by a nonunit pivot, universal four-empty integrality, a leading quantile without a charged block, and an attempt to exclude size 91. The general mathematical reductions are proved here and in the cited foundation; these finite checks are not a proof-assistant formalization.

Only Python 3.10+ and its standard library are required for verification:

```
python -I -S -B verification/upper91.py --output .work/upper91.json
python -I -S -B verification/upper91.py --exhaustive
```

The data are `data/upper91_certificates.json`. The optional `experiments/generate_upper91_certificates.py` uses SciPy to discover candidates; exact integer arithmetic determines acceptance. Standard and optimized Python outputs agree byte for byte. The exhaustive result agrees except for its mode label, regenerated certificates are byte-identical, and the unchanged upper-92 baseline is reproduced. The large standard result is archived as `result.json.gz`; lossless decompression recovers the exact verifier-output bytes. `result_archive.json` records both hashes and the roundtrip check. Run receipts, hashes, center-basis witnesses, scalar data and logs are in `results/2026-10-17/upper91/`. The Windows entry point runs all eleven audited finite continuations. The archived endpoint 84 has not thereby received a complete independent replay.

References

[1] V. F. Lev, *On consecutive subset sums*, Discrete Mathematics 187 (1998), 151–160, Theorem 1. doi:10.1016/S0012-365X(98)80006-X.

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[3] Sum-of-Subset research project, *An integer prefix inverse and the audited bound $N \le 93$*, `reports/upper93_from94_20261016.tex`, 16 October 2026.

[4] Sum-of-Subset research project, *A sharper non-progression lemma and the audited bound $N \le 94$*, `reports/upper94_from95_20261016.tex`, 16 October 2026.

[5] Sum-of-Subset research project, *Joint covering losses and the audited bound $N \le 95$*, `reports/upper95_from96_20261016.tex`, 16 October 2026.

[6] Sum-of-Subset research project, *The audited continuation from 115 to 103*, `reports/upper103_from115_20261014.tex`, 14 October 2026.